

GREENING CITIES REPORT:

Greening the Bibliothèque Historique de la Ville de Paris
(BHVP) and its surroundings

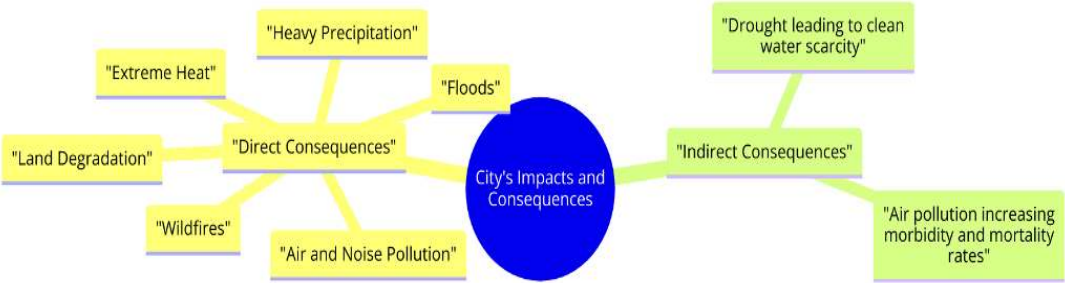
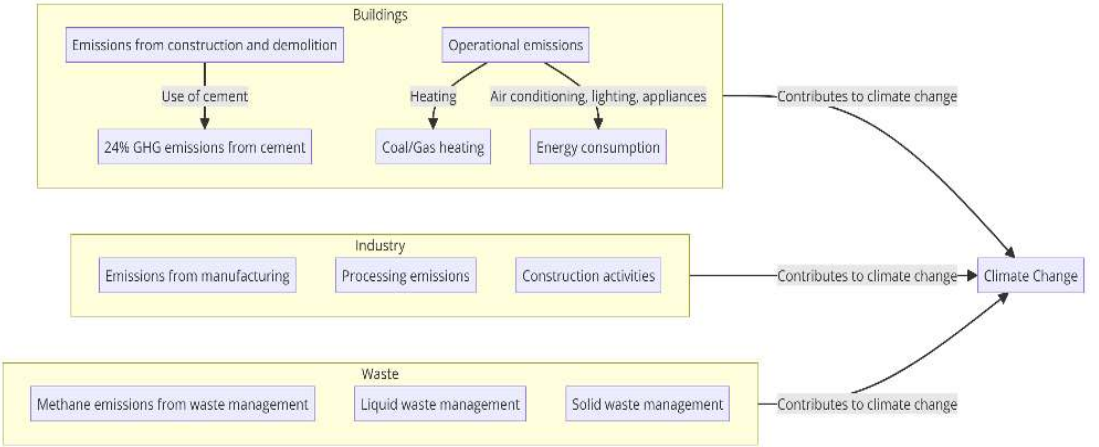
22146303
Bachelor of Arts and Sciences
BARC0172 Greening Cities 2023-24

GENERAL INTRODUCTION

Making cities more green and resilient now constitutes an important part of Local Urban Plans' priorities, especially in high density urban settings like metropolises, which are both responsible for (Cities' activities account for over 70% of global CO2 emissions + see diagram 1), (Weforum, 2023) and have now become victims of climate change and its consequences (see diagram 2). The consequences of this are both direct (floods, air and noise pollution, extreme heat, heavy precipitation, land degradation and wildfires) and indirect (drought that leads to clean water scarcity, air pollution that increases morbidity and mortality rates etc.), (Muckim and Roberts, 2022).

Cities can mitigate these impacts by reshaping their design to prioritize low-carbon solutions, including (re)adapting the built environment and city functioning structures to respond effectively to these risks, notably through the implementation of Green Infrastructure. Green Infrastructure (GI) represents an approach that centres on the enhanced valuation of urban vegetation. This manifests through the integration of natural and semi-natural spaces—including parks, gardens, green corridors, biodiversity and conservation areas—, (IADB, 2022), which can moderate “urban heat island” effects through GHG absorption and reduce the energy consumption required from cooling in buildings as well as improve water management. GI can also help strengthen social ties when applied in public spaces and ultimately improve human health.

This report focuses on the means that can be employed in a targeted area in the heart of the city centre of a big metropole like Paris, France in order to reduce the impacts of the risks posed by climate change and to help mitigate the impacts of the city on climate change.

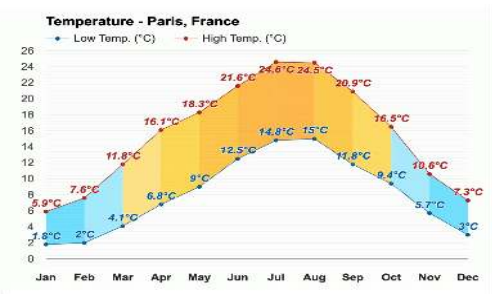


PARIS MACRO ANALYSIS - CLIMATE

Temperatures

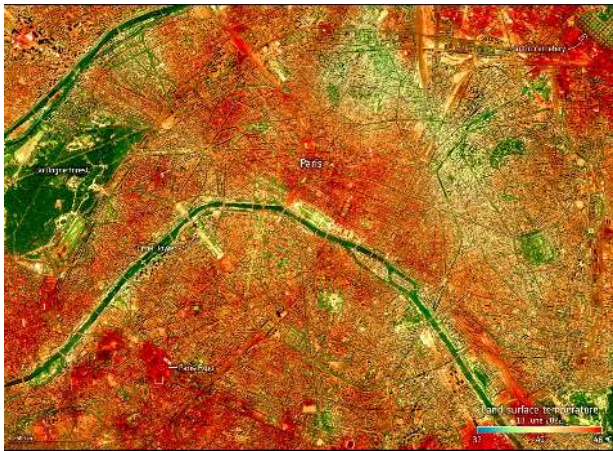
Paris climate is known as oceanic, being comprised in the Cfb Koppen climate classification. This category induces frequent cloudy conditions with precipitations, as well as cool summers with temperatures averaging 22 degrees including infrequent very high temperatures and mild winters. On average, temperatures vary between 1 degree (February) and 24 degrees (September). Yet, in recent years, Paris has suffered from multiple heat waves, notably in 2003, where temperatures reached high records of 43 degrees in multiple cities, and which killed over 15 000 people in a single summer. Since the episode occurred, HVAC systems are proposed as cooling solutions to extreme temperatures and built structures' inability to cool places naturally. However, they emit 117 million tons of GHG per year, and this is expected to only increase (energy central, 2022), as risks of such heat waves that occurred in 2003 to happen again is 10 times higher today, which has been particularly seen in the last 2 summers. This perpetrates a vicious circle. (meteofrance, 2023)

Paris average temperatures and humidity 2023



(d.o.o, 2024)

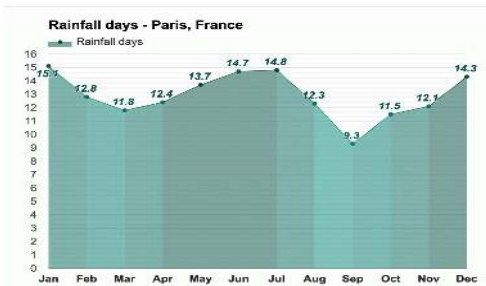
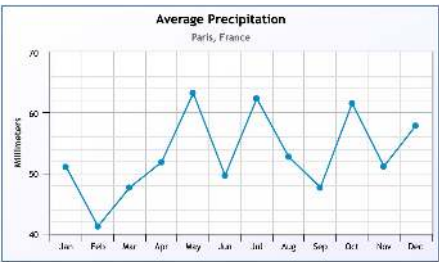
Temperature distribution of Paris-centre during heatwaves



(Weatherbase, 2024b)

Precipitation and rainfalls

Paris has an average annual precipitation of 637 mm (Weatherbase, 2024), mainly in the form of rainfalls, which are distributed evenly throughout the year with peaks occurring in February and December, where average monthly rainfall reaches respectively 41 mm and 49 mm. In addition, on a monthly average, there are more than 1/3 of days with rainfalls.



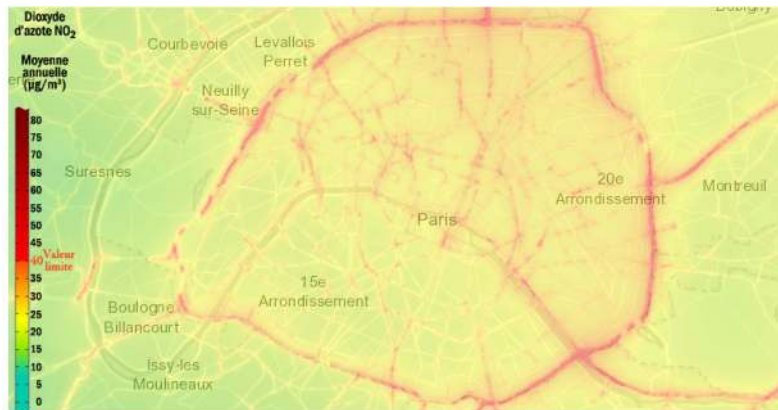
This map shows how temperatures were distributed during the June 2022 heatwave that struck Paris. We can clearly observe a negative correlation between canopy and temperatures.



(Weatherbase, 2024c)

POLLUTION DATA

Pollution can be assessed through the analysis of NO₂ emissions, and PM particles. Both are emitted notably from vehicles and heating systems. Both are extremely harmful to the respiratory system, increasing risks of lung cancer, and asthma attacks, for example (Huangfu and Atkinson, 2020). Reducing the amount of nitrogen dioxide and PM particles presence in the air to the OMS recommendations could have avoided 2,500 deaths in Ile-de-France only in 2019 (Airparif, 2024) . Since 2014, the average concentration of both has been reduced by 40%. We observe high levels of NO₂ next to road axis and in the hyper centre too.



Paris average concentration of NO₂ over the 2023 year



Paris average concentration of PM10 particles over the 2023 year, (Airparif, 2024)

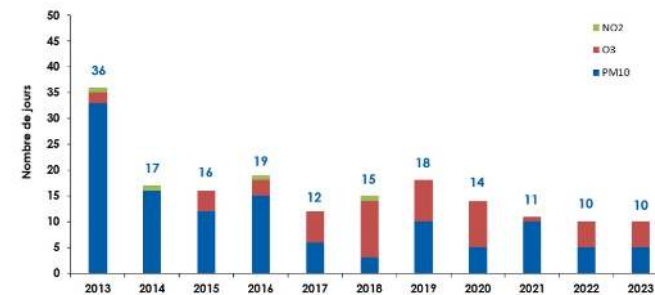
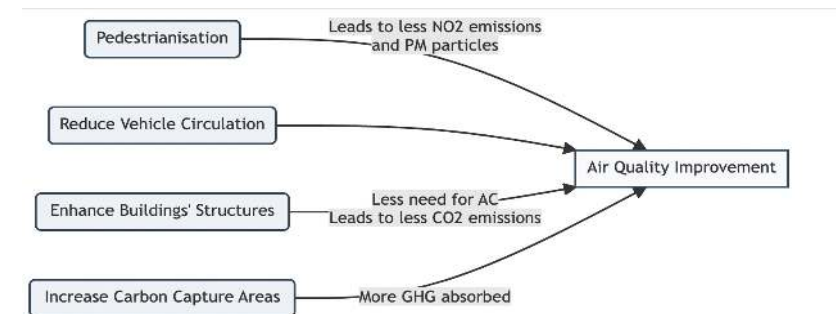


Figure 9 : nombre de dépassement des seuils d'information et d'alerte en Île-de-France de 2013 à 2023

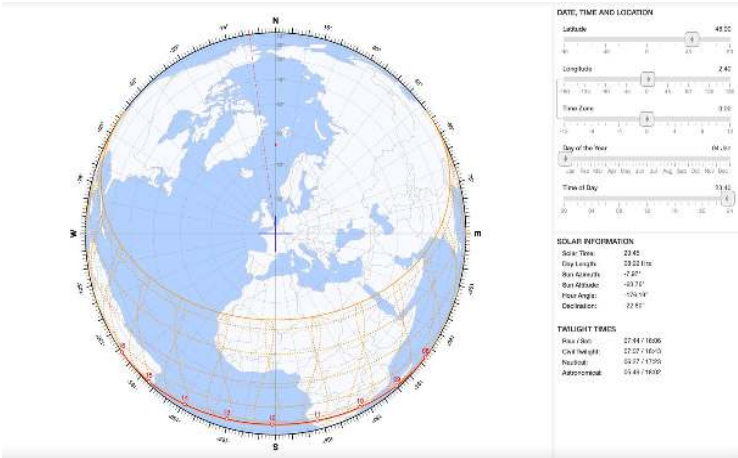
Paris' average number of days per year with levels exceeding the information and alert threshold for each particle (NO₂, O₃, PM10)

Based on these conclusions, it is highly recommended to increase pedestrianisation, reduce the authorised level of circulation of motor vehicles, enhance buildings' structures and efficiencies to decrease the need for air conditioning, increase carbon capture and stockage tools.

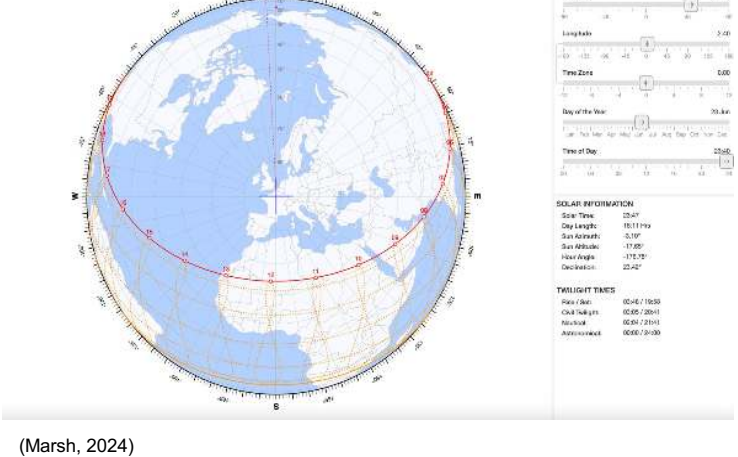


SOLAR ASSESSMENT

Winter



Summer



(Marsh, 2024)

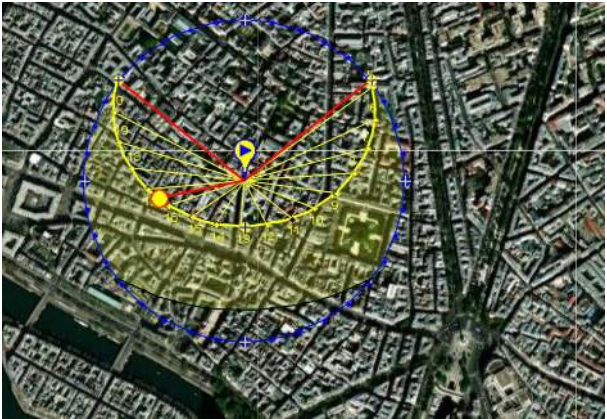
The amount of daylight difference during the shortest day in winter and the longest in summer, are as high as 8 hours (16.15 hours day length during the summer compared to 8.25 hours in the winter.) This can have several implications for the right fit of Green Infrastructure choices on our site. Paris is particularly concerned by the urban heat island effect, and our site especially.

Based on these facts, it is reasonable to say that the implementation of solar panels may be inappropriate due to unequal time of sunlight between seasons as well as the rooves' structures which will be analysed in the next slide. Reasonable GI implementation in response to the site's sun exposure could include: the inclusion of green/white rooves, more ground biodiversity and wall greening, in order to not only benefit from the sunlight provided but also repair the urban heat island effect.

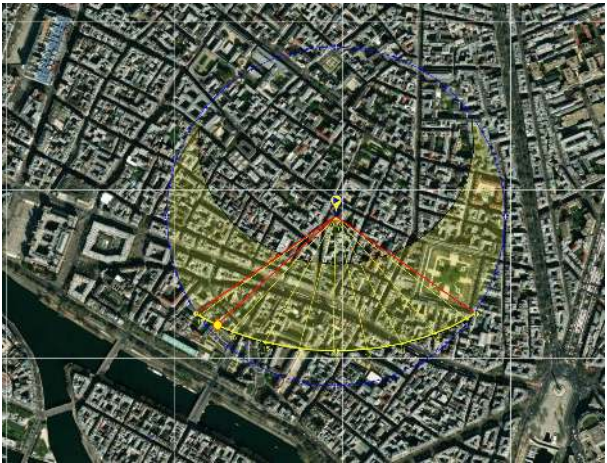


(wozacos, 2015)

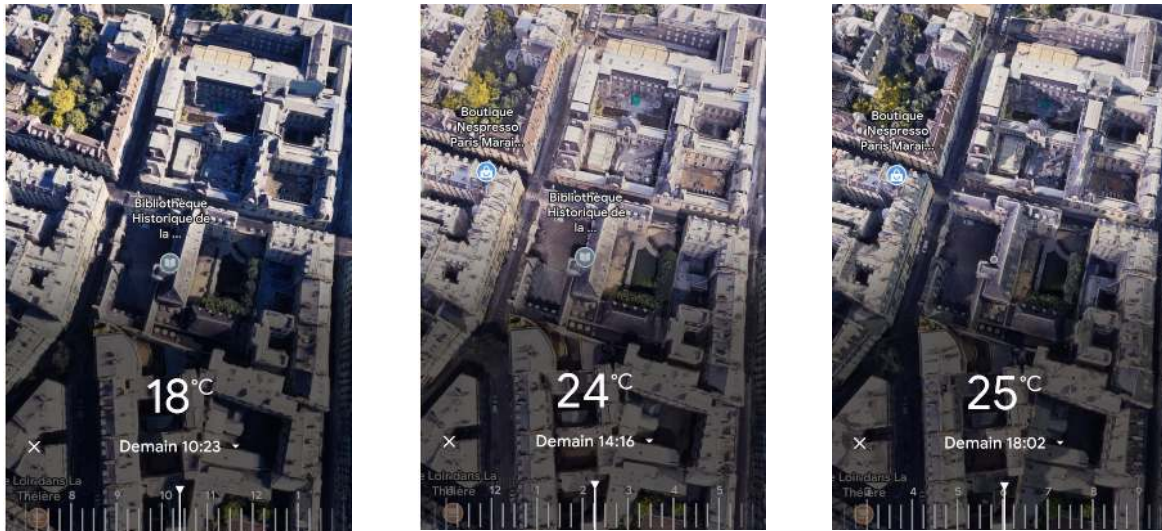
Summer



Winter



FURTHER SOLAR ASSESSMENT



(Google Maps, 2024)

The whole site or part of its courtyards and gardens receive sunlight everyday from 11:00 am to 6:00 pm so for a minimum of 7 hours per day, during a typical Spring day (8th of May). The pick is at 2:00 pm, where all the site is being exposed to sunlight. This can justify the implementation of specific plants relevant to that sun exposure. In contrast, the site's rooves' exposure to sunlight depend on the context.

The site's buildings have rooves from very varying style, including Mansard rooves, especially on the neighbouring buildings of the BHVP, typical of the Quintessential Parisian style of 19th-century architecture style, on which the implementation of rooves is possible. In contrast, the majority of the BHVP's rooves are of hip type, dating from the XVIth century which makes their exposure to sunlight, on each side, split during the day and the application of GI rooves complicated. In addition, some of the site's rooves are relatively flat and below other rooves' levels, making the duration for which their surfaces are exposed to sunlight vary between 7 hours and 10 hours per day (fully exposed in yellow and partly shaded in orange). The potential for green/white roof implementation is labelled in green.



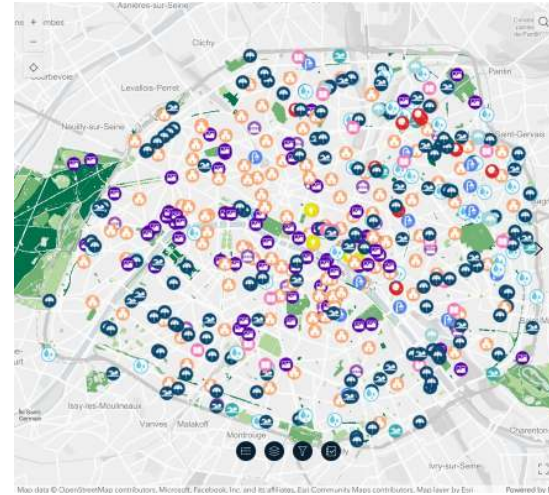
CITY OF PARIS STRATEGIES IN PLACE TO TACKLE CLIMATE CHANGE

Achieved goals:

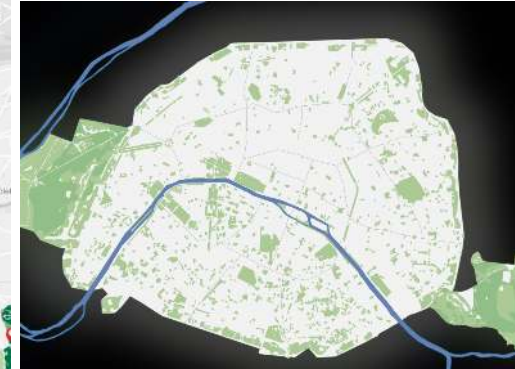
Between 2004 and 2023, local emissions decreased by 36%. This was possible thanks to the implementation of measures targeting the adaptation of residential and tertiary real estate, energy consumption, urbanism, mobility, air quality, feeding, and waste. Notably, by the end of 2023, the city had achieved over 1,300 cool places that serve as places for refreshment during heatwaves, a total of 26.2 % of the territory was vegetalised (with 136 ha of vegetalised roof and walls, including 30 ha of urban agriculture spaces and 167 shared gardens), 1,100 islands of freshness were recensé. (Ville de Paris, 2024). In 2023, Paris has also introduced cool roofs, which are techniques of roof cooling using reflective white surfaces. These traditional roofing techniques can reduce temperatures in rooms below the roof by up to 2 or 3 degrees depending on the conditions. Also, For stormwater zoning considerations, the infiltration of stormwater has been enhanced by debitusising 100 ha of surfaces between 2018 and 2024. (Ville de Paris, 2024)

Future goals:

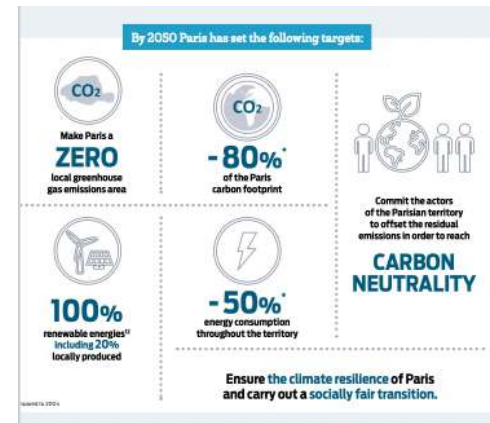
Paris Climate Plan 2024-2030's main goals are to protect the population from climate change consequences, by 1) vegetalising, cooling and adapting the city putting the pedestrian at the heart of projects, in order notably to obtain 10 sq metres of green space per habitant through the implementation of more canopies; 2) reduce inequalities posed by climate change by - recensé and enhancing places that can serve as refuges (fresh public spaces notably) – and helping the transition to low carbon mobilities with the implementation of bicycle parking; 3) massify built structures' environmental transformations through – the implementation of 1000 additional anti-overheating roofs, isolation workings, and reaching 150 ha of vegetalised real estate (Paris, 2024). In addition, whilst 70-80 % of the capital's roofs are still in zinc, contributing to rising building temperatures by absorbing sunlight, multiple private and public initiatives or ideas take place. The mayor has confirmed an objective to equip 20 % of Paris roofs with solar power units by 2050. (Ville de Paris, 2024)



Map of cold islands Paris, (Arcgis, 2023). We observe high concentration of cold islands in the hyper-centre, though this remains an area where levels of NO₂ and PM particules remain high, potentially indicating lack of green spaces, which capture GHG, as shown in the next map.



Map of green spaces, Paris, (sentientt, 2018)



The future of Paris' roofs imagined, (Roofscapes, 2024)

BHVP INTRODUCTION- HISTORY, MAPPING AND ARCHITECTURE. SITE 1

Location zoomed out



Location zoomed in and exact area of work



(Google Maps, 2024)

The chosen site is Bibliothèque Historique de la Ville de Paris (BHVP) and the bits of Rue Pavée, Rue des Francs Bourgeois, and Rue de Sévigné surrounding it, creating an interesting quadrilateral.

BHVP was inaugurated in the late XIXth century and most of its built structure is XVIth century old.



Newly added aisle from the XXth Century **GREEN**



XVIth century aisles, structure and building were kept. **BLUE**.



Mainly kept structure and building from the XVIth century. **YELLOW**



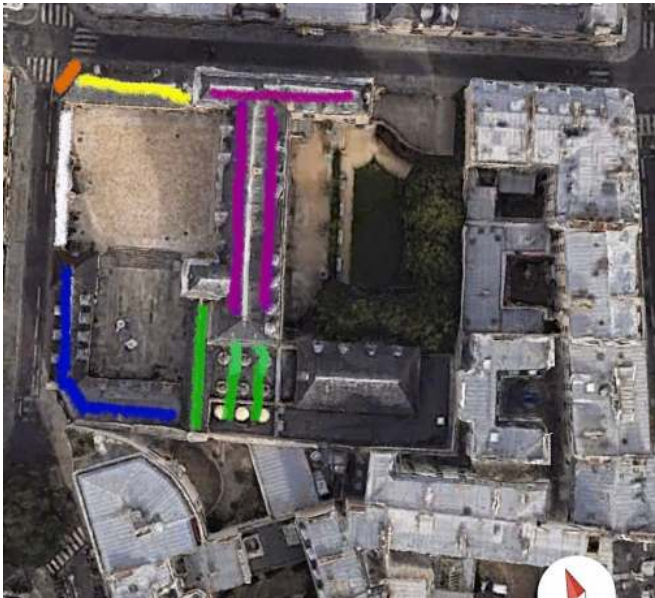
Kept corner turret from 1620. **ORANGE**



Front and back of the building's old structure from the XVIth century kept but renovated and transformed in the late XXth century. **VIOLET**



Entrance portal dating from 1718. **WHITE**



Coded Earth map of the library site

NEIGHBOURING SITES

Jardin de l'Hôtel Lamoignon. Site 2



The Jardin de l'Hôtel Lamoignon (violet area) is a public garden located behind the east wing of the BHVP, with an access Rue des Francs-Bourgeois. Whilst it used to belong to the library, it no longer does.

The garden offers a basic lawn (yellow), inaccessible for visitors to step on. It also has a nice protected curved flower and plants bed (blue). There is a terrasse towards the west of the site (green), whose access to the public is forbidden. It has plants pots and the terrace is exposed to sunlight towards the west. We also observe climbing plants, present all over the neighbouring building façade, on the east side of the garden. The walls are very high and their plants have evident lack of maintenance.

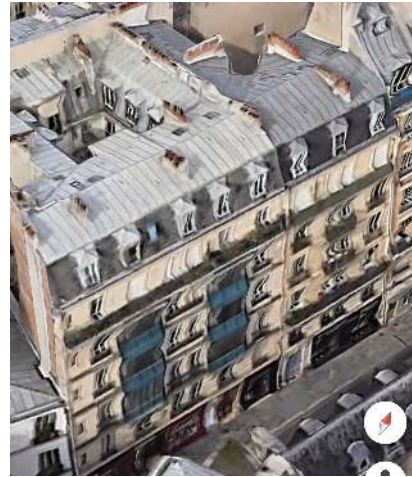


Overall, whilst the area is big, averaging 330 square metres, it is underexploited and restricted in terms of green infrastructure and public access and use. On a ground level, the lawn area could be enhanced by incorporating more biodiversity. In addition, the restricted access to terraces prevents the public from enjoying a very needed outdoor area.

NEIGHBOURING SITES

Rue Pavée: site 3

Rue Pavée is where the BHVP entrance is located. The buildings facing the library are residential and of Haussmannian style. The first two buildings starting from the south (yellow) have Mansard type rooves, in zinc and totally not white painted nor greened. The last two buildings' roofs (purple) are much more complex and superposed. For the latter two buildings, the attic apartments loated on the last floors are directly exposed to heat and sunlight, and hence very sensitive to heatwaves.



Yellow labelled roofs

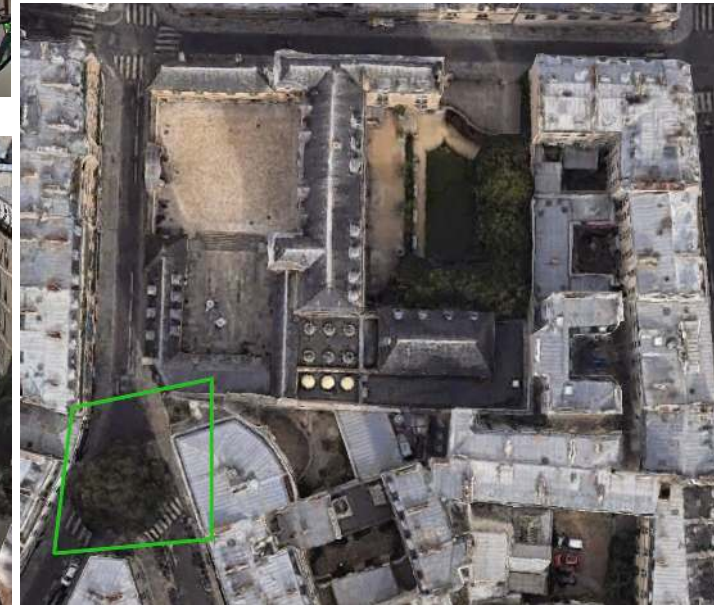


pink labelled roofs



Arbre Micocoulier: site 4

Further down Rue Pavée, is located a s roundabout with a very unique hackberry tree. Heckberry trees are well suited for environments like Paris, being resistant to pollution, heat and tough soils . The potential for pedestrianisation and cycle baking is important, though the area remains relatively small. There is potential for the greening of the building's facade in front of the BHVP. Finally, there is also potential for the greening of the zinc roof of that same building.



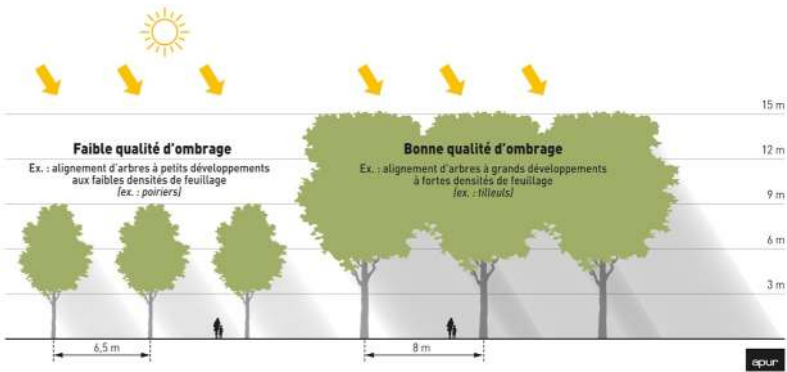
GREEN INFRASTRUCTURE: LITERATURE REVIEW

I) Strategies:

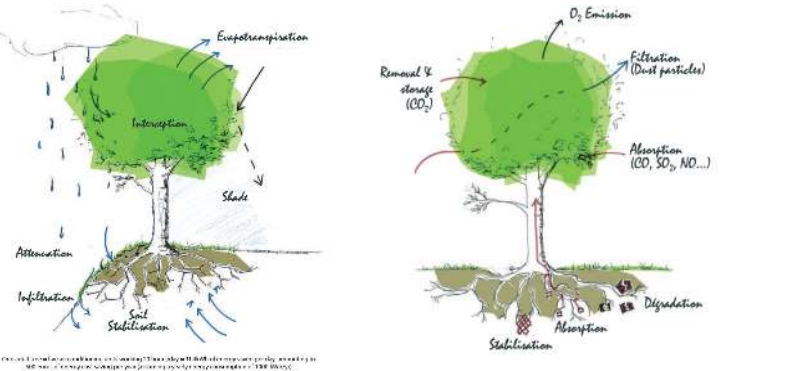
There are three main objectives to be achieved on the environment in highly urbanised areas: the cooling of the site, the pollution emission reduction and enhanced water management.

On a ground level, from a cooling perspective, the first solution can be achieved with the increased vegetation to reduce the warmth effects caused by solar rays. On our site, an effective approach reside in the shadowing of Paris’ streets bitumen surfaces, and in the evapotranspiration of vegetation, in order to reduce urban heat. Bitumen and mineral surfaces absorb an important share of the warmth brought by sun rays., contributing highly to the urban heat island effect. Although the City of Paris has adopted the practice of regularly sprinkling the streets every half-hour in summer to lower temperatures by 3 to 5 degrees, this may seem counterproductive, as it uses an unreasonable amount of water during heatwaves, amounting to half a shower per resident per day should all streets be watered, (Echelard, 2022). There are a number of NBS, both vertical and horizontal to approach the need for increased cooling on street and bulding level, and combining both is crucial, (Agence Qualité Construction, 2024)

Developed trees can reduce up to 80% of sunlight exposure, with species like chestnuts and lindens providing deep, extensive shade, while plane trees, maples, and pagoda trees offer a more sparse canopy. This varied shading is essential for cooling bitumen surfaces that absorb significant heat from solar rays. Furthermore, the layering of vegetation through species diversity enhances ventilation. In narrow areas, such as the ones we are dealing with, line trees are often imposed. Their productivity is constrained by size regulations, and their compact and small soils which are often surrounded by poor biodiversity and bitumen surfaces. To mitigate these effects, the planting of such trees in permeable soils that can infiltrate the water needed for their growth, possible by their proximity to rain gardens, bioswales or planter boxes, for example, is crucial. By increasing the activity of photosynthesis, evapotranspiration and shadowing, thanks to higher canopy, temperatures could decrease by 1 to 3 degrees, (Agence Qualité Construction, 2024). Planting trees in such spaces provides an ecosystem of plants and species that further contribute to that effect, possibly creating a virtuous circle of biodiversity and urban cooling. Challenges involve taking into account the right tree type in the face of people’s allergens and environment.

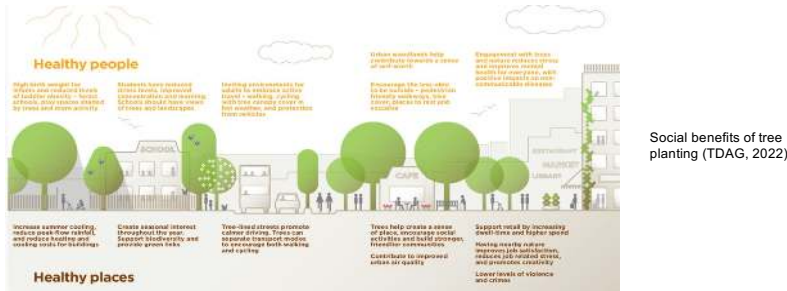


Guide for effective shadowing. "Weak shadowing quality" and "good shadowing quality", (Agence Qualité Construction, 2024)



Explained effectiveness in reducing temperatures (GrandLyon, 2015)

Explained effectiveness in reducing air pollution



CASE STUDIES TREES



Example of tree planting whose efficiency is challenged by the tree's continuous exposure both to sunlight and artificial light because of the floor lamp presence (Pinterest, 2023)



Example of narrow street with effective street trees implementation, with enough interspace between them and a solid plantation foundation that can feed the tree with appropriate water levels and biodiversity thanks to the irrigation properties of the small garden and the diversity of the plants. Additional natural filters can be added in the soils to increase the water absorption of the plants and trees, such as charcoal.



Tree pits:
(GreenBlueUrbanLtd, 2024)
The Stockholm tree pit effectively collects water and supports layered vegetation,



Community involvement in tree planting



(ICE, 2024) Effective line forest of trees aligned, only on one side of the street, to avoid pollution to be retained

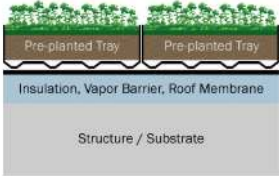
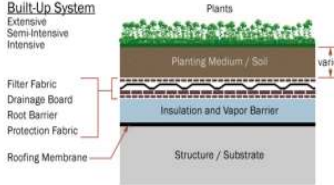


(GreenBlue, 2024) Tree planting effective against drought

GREEN INFRASTRUCTURE LITERATURE REVIEW

Green infrastructure should also be implemented on roof level, as briefly introduced earlier, as buildings’ heat also comes from the reflection of sun rays on roofs, especially when made of zinc, which stocks solar energy a lot, contributing to rising temperatures within ~~buildings~~ and flats (Arup, XX). It is reported that vegetalised rooves reduce energy consumption, air conditionning as well as heating in important ways, reaching levels of 35 % less energy consumption in certain cases (Canada study: (Apur, 2022))

In addition, green roofs can provide a habitat for biodiversity and species as well as a create outdoor spaces for residents. They also have the capacity to reduce the volume of stormwater runoff by evapotranspiration and absorption of the vegetation. In Paris, their implementation in crucial, as 80% of the city’s roofs are zinc-made. There are multiple type of green roofs each with different implication.

 			
Extensive Green Roof Tray System Built-up Green Roof System: Semi-intensive or Intensive			
Type of Green Roof	Environmental Benefits	Management Aspects	Socioeconomic Benefits
General for all types	- Stormwater quantity and flow reduction		- Local job creation
	- Reduction in urban heat island effect		- Raising property values
	- Improved air quality		
Extensive Green Roof	- Lightweight, suitable for large areas	- Low maintenance: no irrigation needed	- Low cost, easy to replace
	- Reduces volume of stormwater runoff	- Vegetated with low-growing, drought-tolerant vegetation	- Brings nature into the city
		- Retention: 0.5-1.5 gal/ft²	
Semi-Intensive Green Roof (Hybrid)	- Combines features of both intensive and extensive roofs	- Periodic maintenance	- Greater coverage at relatively low cost
	- Greater plant diversity due to deeper soil	- Periodic irrigation	- Opportunities for plant diversity education
		- Retention: 1-2.5 gal/ft²	- Enriching biodiversity knowledge and increasing sensitivity to sustainability issues
Intensive Green Roof	- Largest plant diversity	- Regular maintenance and irrigation	- Usually accessible
	- Best for insulation and stormwater management	- Supports lawns, perennials, shrubs, trees	- Greatest variety of use
	- Greater retention: 2-5 gal/ft²	- Potential for urban food growing	- Building tenant participation in farming/harvesting activities, creating a sense of community



The scheme illustrates in a simple manner the impact of green roof implementation on temperatures and water management too. First, they enhance hydrologic performance by restoring natural processes such as infiltration, storage, and evapotranspiration, thus reducing the volume of stormwater runoff. Studies show that green roofs can retain up to 80% of annual rainfall, depending on their structure and the local climate, which substantially mitigates the volume of water flowing into storm drains. Second, they provide a buffer that delays and reduces peak flow rates during rainfall events. By detaining stormwater, green roofs slow its entry into the drainage system, thereby preventing the overwhelming of urban sewer systems and reducing the risk of flooding. Lastly, the layered structure of green roofs, including soil and vegetation, acts as a natural filter, improving the quality of stormwater by adsorbing pollutants before they reach watercourses. This capability positions green roofs as a valuable component of sustainable urban drainage systems, contributing to cleaner urban waterways, (Palla and Gnecco, 2018)



Extensive green roof example



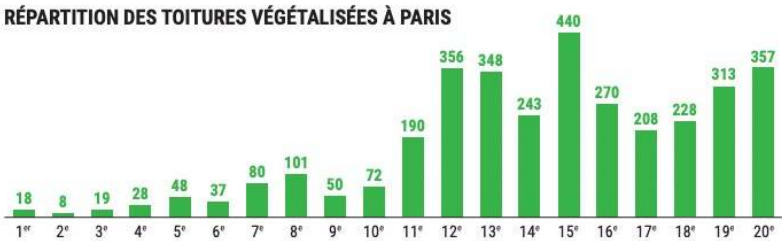
Semi intensive green roof example, greater biodiversity



Intensive green roof example and community engagement

CASE STUDIES GREEN ROOFS

RÉPARTITION DES TOITURES VÉGÉTALISÉES À PARIS



Paris green roof repartition according to areas, there is an evident under representation of green roofs in the 4th district, where our site is located.

Example of partially vegetalised zinc rooftops, Paris 4th and 5th arrondissements.



Green roof addition ideas by Roofscapes Studio startup, (Roofscapes, 2024)



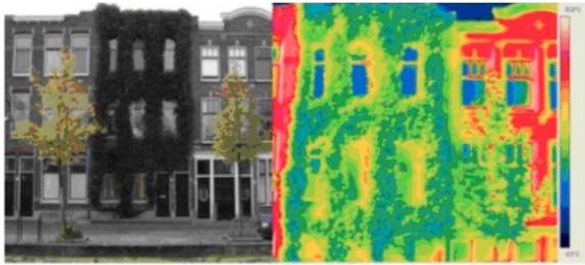
Example of rooftop involving community engagement for planting and maintenance on flat surfaces



(Apur, 2022b)

GREEN INFRASTRUCTURE LITERATURE REVIEW

Green walls are also a tool that can be used to protect a building's structure from heat effects, as temperatures behind green walls can be reduced by as much as 10 degree and stormwater effect, hence saving energy expanses as well as purifying air by absorbing GHG. Green walls can help reduce sound reflection too (Asla, 2022)`; Through shading, green walls can lower temperatures in summer and reduce energy costs by 23 percent. Their benefits are similar to that of green roof. In terms of heat, Green walls contribute to cooling buildings through the evapotranspiration of plants, and their shading property on the building facade. This natural cooling effect helps mitigate the urban heat island effect, reducing the need for air conditioning, leading to energy savings and decreased emissions from power plants. There are several types of green walls: living walls, which are particularly advanced types of green walls, that incorporate systems that can manage water efficiently. They often use hydroponic systems that circulate water, reducing the need for additional watering. This not only conserves water but also captures and recycles rainwater, which decreases the burden on urban drainage systems by reducing runoff (Manso and Teotónio, 2021). Moreover, the presence of plants helps filter and purify this water naturally, which can lead to improved water quality before it is discharged back into the urban environment or reused within the building. Despite these advantages, the implementation of living walls involves challenges, primarily related to cost. The initial setup, which includes specialized frames and irrigation systems, can be expensive. Moreover, living walls require ongoing maintenance such as nutrient replenishment and plant care, which incurs important costs. It is also important to choose a right set of plants that survive winter seasons in face of strong winds, dry air and snow falls, which can happen in Paris. These factors make living walls a significant investment compared to other greening solutions and require important maintenance costs. There are other type of green wall solutions including modular panel systems and trellis systems. Modular panel systems offer a more cost-effective and less complex alternative, providing structured environments for plant growth that can moderately absorb rainwater and assist in water management. Trellis systems, while less capable in water management, effectively mitigate urban heat by shading building facades, thus contributing indirectly to reduced energy use in cooling systems. These alternatives each offer unique benefits, allowing for flexibility in addressing specific urban environmental challenges.



Green wall's cooling impact on building temperatures,
<https://icap.sustainability.illinois.edu>

CASE STUDIES



L'Oasis d'Aboukir, a 25 meter-high green wall featuring over 250 species



Living walls implemented by Patrick Blanc, projects in Paris

<https://www.verticalgardenpatrickblanc.com/node/4676>

SERRES D'AUTEUIL CASE STUDY

Located in the South-West of Paris, right behind Paris famous ring road, the “periphérique”, this garden offers an invaluable amount of plants, trees, biodiversity and greenhouses to the area. Doing on-site research there gave unique advice and real case studies of what kind of right nature there is to implement in an urbanised context.

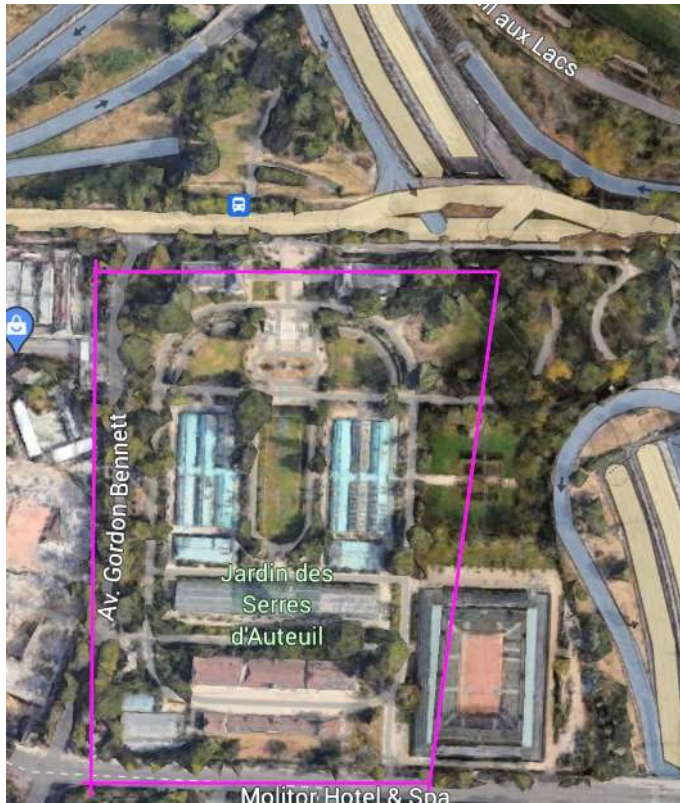


Photo 1



Photo 2



Photo 3



Photo 4

Photo 1 and 3: Unique design for effective water management, incorporating both gratings and vegetation.

Photo 2: Raised bed set up with the presence of very diverse plants and flowers and palm trees. Photo 4: Idea of curved shaped raised bed incorporating plants and flowers, being relatively vertical too.

Photo 5: Example of community garden where plants and vegetables are being grown, protected by a small barrier.



Photo 5

RAIN WATER MANAGEMENT LITERATURE REVIEW AND EXAMPLES

Bioswale



Rain gardens for green roofs



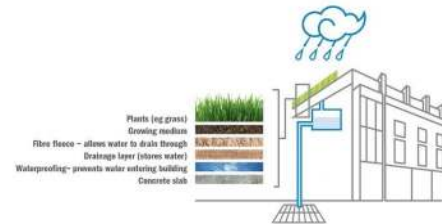
Permeable paving infiltrating water



Blue green roofs



Blue green roof functioning



Stormwater planters, (Water Phila, 2023)



In urban environments, managing rainwater efficiently is crucial to mitigate the risk of flooding and to conserve water resources. Traditional drainage systems can become overwhelmed during heavy rainfalls, as water from impervious surfaces like roofs and pavements rushes into sewers, potentially leading to severe environmental and infrastructural problems (Denchak, 2022).

A 'blue-green roof' incorporates elements of both greenery and water management. It is able to collect and store rainwater temporarily, delaying its journey to the sewer system. By releasing stored water slowly into the drains, blue-green roofs can significantly reduce the risk of overwhelming city drainage systems and mitigate flood risks (Kasmi, Bruel and Nappi, 2023). The captured rainwater can be reused for non-potable purposes such as irrigation and toilet flushing, significantly reducing the demand on municipal water supplies.

Bioswales are linear channels designed to concentrate and convey stormwater runoff while removing debris and pollution. They are typically vegetated with a mix of grasses and perennials that can tolerate both wet and dry conditions. The plants and soil in bioswales filter pollutants from the water by trapping sediments and absorbing nutrients and contaminants, such as phosphates and heavy metals, thereby reducing pollution entering waterways.

Rain gardens are shallow depressions planted with native shrubs, and flowers that are positioned near runoff sources to capture rainwater from surfaces like roofs, driveways, and lawn areas. The design allows rainwater to soak into the ground naturally, decreasing the volume and flow rate that reaches sewers and streams, thus mitigating flood risks. By mimicking the natural hydrologic cycle, rain gardens enhance the local microclimate and provide habitats for various bird species, thereby supporting local biodiversity.

CASE STUDIES

Example of permeable pavements, facilitating water infiltration



building bases planted, (Apur, 2020)



Urban community farming.



Compost garden beds



Planter boxes



Green roof bike shelters



(Cameron, 2024)

Green roof sun shelter



Community tree planting



(Cornell Cals, 2022)



Urban bee hives on green roofs, providing homemade honey for residents. (Pinterest, 2022)



(Cycle & Bin Shelters, 2024.)

SITE 1 PROPOSAL

- BHVP has an outside area of 625 SQM of courtyard (including 400 squared metres of pavement from the lower paved courtyard and 225 square meters from the upper courtyard). Fully making the ground permeable again is possible for 75% of the site, less than 30 % of the site is vegetelised, falling in the lowest category. (Apur, 2020b)

Proposal:

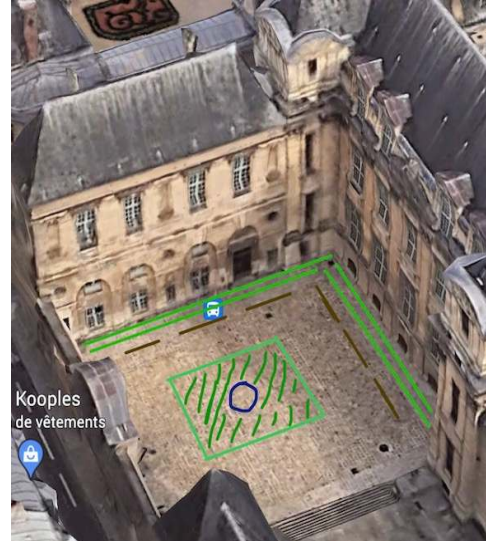
Lower courtyard (scheme 1 and 2):

- Make all pavements permeable and grassed (see page 18 for illustration)
- Implement at least 2 green roof bike shelters on each side of the portal.
- Planting at the base of every building (through raised beds or planter boxes if not applicable, see page 18 for illustration) and implementation of living walls on both walls surrounding the portal (see page 15).
- Adding of benches (brown traits) and of a central biodiversity garden in a French garden style.

Upper courtyard (scheme 3):

- Implementation of compost garden beds and planter boxes along all buildings' facades' bases (see page 18): objective: provide a sense of belonging to the students coming to the library and making them participate in the maintenance of the area. Also visual interest, with biodiversity impacting mental health positively.
- Adding of green roof sun shelters (see page 18): in the summer to protect from the sun whilst still enjoying some cooled outdoor space, and during rainy days to be protected. Add benches too for increased occupation capacity (black traits).

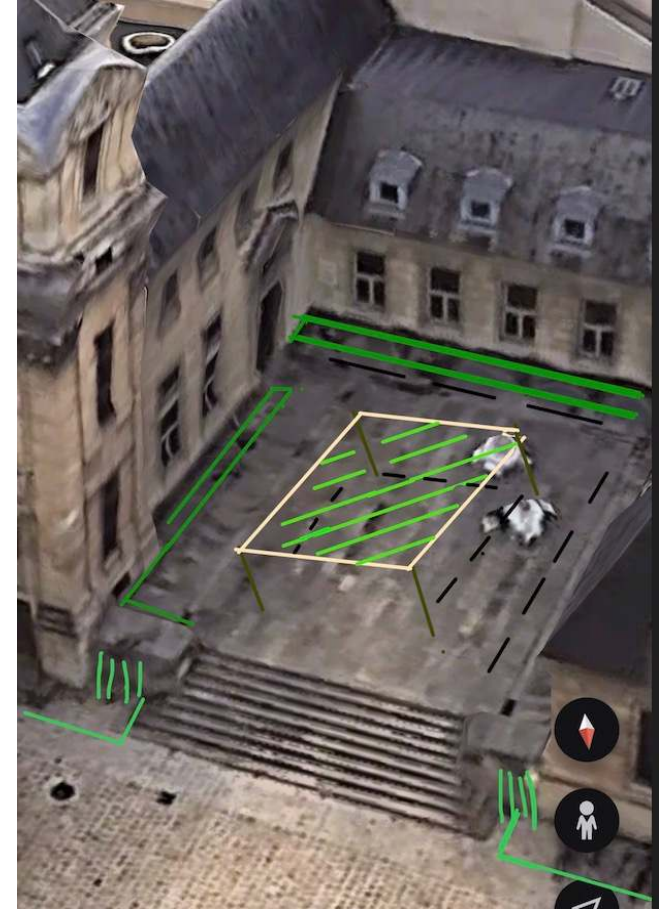
Scheme 1



Scheme 2



Scheme 3



SITE 2 PROPOSAL

- Implementation of green wall from the bottom to the top with a variety of plant species (excluding allergens one) along all the site's south walls.
- Transformation of the lawn into two divided urban community gardens / local plantation accessible to the public to visit by creating a pathway in the middle. Potential participation from local residents and local primary schools to the gardening and harvest during defined periods of week days under the supervision of a gardener.
- Implementation of low maintenance planter boxes all along the walls' origins.
- Change of the flat surfaces on the south block's (public building) and east block's (residential building) roofs into semi-intensive green roofs, to prevent hot temperatures within the building during the summer and retain heat during the winter, and to better evacuate rain water (slide 13 scheme).
- Transformation of the curved plant bed into another one incorporating much more plant diversity and favouring species development
- Access of the west-wing terrasse to the public



Personal picture



Green wall redone, (before, after)



Curved before before and after



Jardin des Serres d'Auteuil



Planter boxes (Pinterest, 2024)



Alamy, 2024

Lawn transformed into an urban community garden, (before, after)



Transformation into semi-intensive green roofs

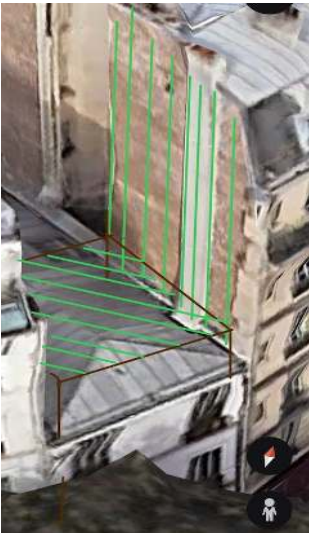
SITE 3 PROPOSAL



Design inspiration from the StartUp looking to green Paris' zinc roofs, with urban hives and public access. (Roofscapes, 2024)



Implementation of cycle paths(white dark lines) instead of current paths for cars with the adding of a line forest of trees(ex. Stockholm treep aligned and bioswales an stormwater plants (green lines) (see page 12 for illustration) in the middle, crucial in this area massively lacking of trees. Size of sidewalks unchanged. Adding of pedestrian crossings on each side to make the roundabout liveable for pedestrians with benches (represented by light white traits)



Wall greening of the the building's façade and adding of another green roof terrasse

SITE 4 PROPOSAL



(Blanc, 2013)

Wall greening of the façade (before and after). Inspiration from Patrick Blanc's workings on greening Paris' city walls

Implementation of solar panels: black lines, and of extensive green roofs (green lines) incorporating rainwater gardens too. Praticle because it is a surface constantly exposed to sunlight, with a high potential of greening (Apur, 2020).

Design inspiration. Important to note that this is subject to debate, aware of the disparity of available daylight between seasons.



Extensive green roof inspiration



(netconstructions.de, 2020)

SUMMARY OF THE PROPOSAL'S CONSIDERATIONS

Involved Parties	Site 1	Site 2	Site 3	Site 4
Design and Planning	- Paris City Architects - Exterior Furniture Designers - Urban Planners - Experienced Project Professionals	- Paris City Architects - Exterior Furniture Designers - Green Roof Specialists (cost-effective choices) - Green Wall Designers - Horticulturists to select optimal plants	- Guidance from the local mayor - Urban Design and roof design Start-ups - Professional Gardeners for tree planting - Architects and urbanists for bicycle lanes - Arborists to select optimal tree planting	- Architects - Solar Panel Specialists - Green Wall Designers
Local Community	- Engage local residents - Involve site employees - Public input on green area design (library as public space)	- Collaboration between the city and building residents - Community consultation on design and cost-sharing - Public participation in project implementation	- Engage local residents - Public consultation for street transformation to bike paths - Negotiation on car route adjustments	- Approval from building residents - Negotiation for green roofs and solar panels - Community input on green wall maintenance
Wildlife Groups	- Guidance from urban horticulturists - Selection of climate-resilient plants - Wildlife integration advice	- Guidance from urban horticulturists - Selection of appropriate plant species - Advice on local wildlife integration (e.g., bees, birds)	- Guidance from wildlife experts on bees hives - Safety considerations for species - Plant species selection for rooftop gardens	- Guidance from wildlife experts - Suitable plant species for green walls and roofs - Integration of beneficial wildlife - Selection of optimal tree species
Employment	- Job creation in gardening, maintenance, and security - Possible increase in library staff - Community and school participation in maintenance	- Job creation in gardening and maintenance - Potential additional staff for increased green areas - Cost-sharing among residents	- Job opportunities in construction, gardening, and maintenance - Potential for community-based maintenance programs	- Job opportunities in green wall and solar panel installation and maintenance - Potential for community-based maintenance
Education	- Educational programs on biodiversity and sustainability - Workshops for local schools and community groups	- Educational initiatives on green roof and green wall benefits - Workshops and community engagement activities in urban farms and community gardens	- Educational programs on urban biodiversity - Community workshops on green roof maintenance and benefits	- Educational programmes on green wall and roof benefits - Community engagement and sustainability workshops for building residents
Health Sectors	- Health benefits from increased green spaces - Programs promoting physical and mental well-being through green areas	- Increased green space per resident - Initiatives promoting community well-being	- Health benefits from reduced pollution and heating and increased green areas potentially accessible for leisure - Programmes promoting active lifestyles with new bike paths	- Health benefits from urban greening and improved air quality, cooling and energy saving from solar panels
Construction	- Implementation by professional construction firms - Urban gardening experts for green areas - Cost considerations for the city	- Green roof and green wall installation by specialized firms, potential collaboration with startups - Cost analysis for green roof and green wall types - Collaboration with building residents for cost-sharing	- Street transformation to bike paths - Coordination with residents and city authorities - Green roof construction with input from startups and architects	- Green wall and roof installation by specialised firms - Negotiation for solar panels - Cost and benefit analysis

CONCLUSION AND LIMITATIONS

This report focused on transforming a relatively small yet historically rich area around BHVP, incorporating both public and private held spaces into a model of urban green infrastructure. Despite the limited space, the proposals detailed here ambitiously integrate a wide array of green elements—from bicycle lanes to wall greening and green roofing to redesigning the courtyards of the library with specific vegetation. These elements are designed not only to enhance the local environment but also to align with broader city goals under Paris's climate strategy, that we have detailed too.

While ambitious, the scope of greening in such a prestigious and constrained setting presents clear challenges. The historical significance of the site restricts extensive modifications, and the integration of green technology requires careful planning to balance conservation with innovation. In addition, there would be difficulties in reuniting all residents' opinions in one consensus, because of the magnitude of the change, though it remains in a relatively small area, and because no such project has been done before. Moreover, the financial and maintenance considerations of such projects, particularly in densely built historical centres, cannot be overlooked and would come at the costs of both public and private parties in our very project.

Nevertheless, this project demonstrates that even small urban areas can contribute significantly to a city's sustainability objectives. By maximizing the green potential of every square meter, the project underlines the impact of thoughtful, integrated urban planning. This initiative not only promises to enhance the quality of life and ecological health of the Parisian urban landscape but also could position Paris to further cement its reputation as a leader in global urban sustainability efforts. The approach taken here could inspire other cities to undertake similar enhancements in small, historical areas, proving that size does not constrain environmental potential.

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